

Analysis of Optimization Methods via non-convex Performance Estimation

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Goal : Analysis of optimization methods

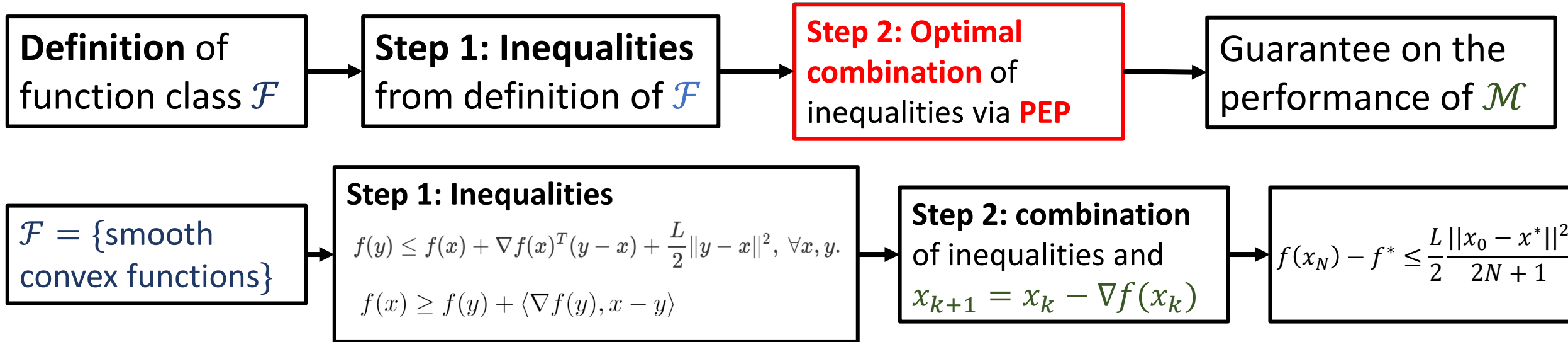
- Optimization method $\mathcal{M}$ (e.g. gradient, Newton methods,...)
- Function class $\mathcal{F}$ (e.g. convex, smooth, self-concordant,...)
- Problem : $\min_x f(x)$

Question : Worst-case performance of $\mathcal{M}$ on instance of $\mathcal{F}$?

Example: Worst-case performance of **Gradient Method** on L -smooth convex functions after N iterations?

$$f(x_N) - f^* \leq \frac{L}{2} \frac{\|x_0 - x^*\|^2}{2N + 1}$$

Constructing a proof of convergence rate



Two sources of (possible) conservatism on the guarantee:

- 1) **Inequalities** do not describe exactly the class (and allow « undesired functions »);
- 2) The **combination** is not optimal.

Optimal combination of exact inequalities leads to exact/tight/unimprovable worst-case analysis

Outline

1. Performance Estimation Problem (PEP) Framework

2. Non-convex PEP for second-order methods

3. New convergence results

Conceptual PEP: maximizing the worst-case performance

Idea: Finding the worst-case performance as an optimization problem

$$\max_{x_0, x^*, f}$$

$$\text{Perf}(x_N, f)$$

$$f \in \mathcal{F}$$

$$x_N = \mathcal{M}(x_0, f)$$

$$||\nabla f(x^*)||^2 = 0$$

$$||x_0 - x^*||^2 \leq 1$$

- Maximize Perf of $\mathcal{M}$ among the set of functions $f \in F$
- Perf(x_N, f) can be : $||x_N - x^*||$, $||\nabla f(x_N)||$, $f_N - f^*$

Issue: Untractable since optimization in function space

Solution: Discretizing function f (w.l.o.g. by black-box property of optimization methods)

From conceptual PEP to tractable PEP (1)

Example: Worst-case performance of gradient method on L -smooth convex functions

$$\begin{aligned}
 & \max_{\text{points } x_i, x^*, \text{function } f} && f(x_N) - f(x^*) \\
 & \text{s.t.} && f \text{ } L\text{-smooth convex,} \\
 & && x_{i+1} = x_i - \frac{1}{L} \nabla f(x_i), \\
 & && \|x^* - x_0\|^2 \leq 1, \\
 & && \|\nabla f(x^*)\|^2 = 0.
 \end{aligned}$$



$$\begin{aligned}
 & \max_{\text{points } x_i, x^*, f_i, f^*, g_i, g^*} && f_N - f^* \\
 & \text{s.t.} && \exists f \text{ } L\text{-smooth convex : } \quad f(x_i) = f_i, \quad \nabla f(x_i) = g_i, \\
 & && \quad \quad \quad f(x^*) = f^*, \quad \nabla f(x^*) = g^*, \\
 & && x_{i+1} = x_i - \frac{1}{L} g_i, \\
 & && \|x^* - x_0\|^2 \leq 1, \\
 & && \|g^*\|^2 = 0.
 \end{aligned}$$

Key concept: necessary and sufficient interpolation conditions

Interpolation conditions

Theorem 1: f is L -smooth convex if and only if for all $x, y \in R^n$

$$\begin{aligned} f(y) &\leq f(x) + \nabla f(x)^T(y - x) + \frac{L}{2}\|y - x\|^2, \quad \forall x, y. \\ f(x) &\geq f(y) + \langle \nabla f(y), x - y \rangle \end{aligned}$$

Theorem 2: f is L -smooth convex if and only if for all $x, y \in R^n$

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{1}{2L}\|\nabla f(y) - \nabla f(x)\|^2, \quad \forall x, y.$$

Proof/PEP does not use all $x, y \in R^n$, only $x_0, \dots, x_N, x^*$

Theorem 3: Given $\{(x_1, g_1, f_1), \dots, (x_N, g_N, f_N)\}$,

$\exists$ L -smooth convex f such that $f(x_i) = f_i$ and $\nabla f(x_i) = g_i \quad \forall i$, if and only if

$$f_j \geq f_k + g_k^T(x_j - x_k) + \frac{1}{2L}\|g_j - g_k\|^2, \quad \forall j, k.$$

From conceptual PEP to tractable PEP (2)

Example: Worst-case performance of gradient method on L -smooth convex functions

$$\begin{aligned}
 & \max_{\text{points } x_i, x^*, f_i, f^*, g_i, g^*} f_N - f^* \\
 & \text{s.t.} \quad \exists f \text{ } L\text{-smooth convex : } f(x_i) = f_i, \quad \nabla f(x_i) = g_i, \\
 & \quad \quad \quad f(x^*) = f^*, \quad \nabla f(x^*) = g^*, \\
 & \quad \quad \quad x_{i+1} = x_i - \frac{1}{L} g_i, \\
 & \quad \quad \quad \|x^* - x_0\|^2 \leq 1, \\
 & \quad \quad \quad \|g^*\|^2 = 0.
 \end{aligned}$$



$$\begin{aligned}
 & \max_{\text{points } x_i, x^*, f_i, f^*, g_i, g^*} f_N - f^* \\
 & \text{s.t.} \quad f_i \geq f_k + g_k^T (x_i - x_k) + \frac{1}{2L} \|g_i - g_k\|^2, \\
 & \quad \quad \quad x_{i+1} = x_i - \frac{1}{L} g_i, \\
 & \quad \quad \quad \|x^* - x_0\|^2 \leq 1, \\
 & \quad \quad \quad \|g^*\|^2 = 0.
 \end{aligned}$$

- Non-convex Quadratically Constrained Quadratic Problem (QCQP)
- Linear on f_i and $x_i^T g_i, x_i^T x_j, g_i^T g_j$
- It can be formulated as convex semidefinite program efficiently solvable !
- PEP gives the exact worst-case numerically (which helps to prove it analytically) [Drori, Teboulle 14]
- It gives all the answers, but we should ask the relevant questions [Taylor, Hendrickx, Glineur 17] 6

Convex formulation of PEP

Convex formulation of PEP when:

Only First-Order methods

- Method analyzed is linear combination of (previous or future) gradients g_i and iterates x_i .
- Interpolation conditions are convex in f_i and $x_i^T g_i$, $x_i^T x_j$, $g_i^T g_j$

1. Gradient method : $x_{i+1} = x_i - \frac{h}{L} \nabla f(x_i)$

OK

2. Fast gradient method :

$$y_{i+1} = x_i - \frac{1}{L} \nabla f(x_i)$$
$$\theta_{i+1} = \frac{1 + \sqrt{4\theta_i^2 + 1}}{2}$$
$$x_{i+1} = y_{i+1} + \frac{\theta_i - 1}{\theta_{i+1}} (y_{i+1} - y_i)$$

See more examples in « PEPit's documentation »

3. Proximal method: $x_{i+1} = \text{prox}_{f(\cdot)}(x_i) = x_i - \nabla f(x_{i+1})$

4. Chambolle-Pock method:

$$\begin{cases} x_{i+1} &= \text{prox}_{\tau f} (x_i - \tau M^T u_i), \\ u_{i+1} &= \text{prox}_{\sigma g^*} (u_i + \sigma M(2x_{i+1} - x_i)), \end{cases}$$

[Drori, Teboulle 14]

[Taylor, Hendrickx, Glineur 17a]

[Taylor, Hendrickx, Glineur 17b]

[B, Hendrickx, Glineur 23]

No Convex formulation of PEP

Convex formulation of PEP when:

Only First-Order methods

- Method analyzed is linear combination of (previous or future) gradients g_i and iterates x_i .
- Interpolation conditions are convex in f_i and $x_i^T g_i$, $x_i^T x_j$, $g_i^T g_j$

1. Newton method: $x_{i+1} = x_i - [\nabla^2 f(x_i)]^{-1} \nabla f(x_i)$ ([de Klerk, Glineur, Taylor 2020] did it for one step)

KO

2. Finite differences : $x_{i+1} = x_i - \frac{f_j - f_i}{x_j - x_i}$

3. Adaptive methods: $\lambda_k = \min \left\{ \sqrt{1 + \theta_{k-1} \lambda_{k-1}}, \frac{\|x^k - x^{k-1}\|}{2\|\nabla f(x^k) - \nabla f(x^{k-1})\|} \right\}$
 $x^{k+1} = x^k - \lambda_k \nabla f(x^k)$
 $\theta_k = \frac{\lambda_k}{\lambda_{k-1}}$ [Malitsky, Mishchenko 2020]

It seems impossible to formulate these PEP in a convex way

Non-Convex formulation of PEP

Idea: Tackle the non-convex formulation of PEP

[Das Gupta, Van Parys, Ryu 2022]

- Analysis of (almost) any method is possible
- Heavy computational cost (global branch and bound solver)

$$\begin{aligned} \max_{\text{points } x_i, x^*, f_i, f^*, g_i, g^*} \quad & f_N - f^* \\ \text{s.t.} \quad & f_i \geq f_k + g_k^T(x_i - x_k) + \frac{1}{2L} \|g_i - g_k\|^2, \\ & x_{i+1} = x_i - \frac{1}{L} g_i, \\ & \|x^* - x_0\|^2 \leq 1, \\ & \|g^*\|^2 = 0. \end{aligned}$$

- Solve the non-convex (QCQP)
- We do not avoid « **Step 1** », we still need a good description of the class considered
- Integer variables and non-quadratic constraints also possible

Idea introduced in [Das Gupta, Van Parys, Ryu 2022] to design methods and used in [Das Gupta, Freund, Sun, Taylor, 2023] to analyze nonlinear conjugate gradients methods.

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Analysis of Second-order methods via Non-Convex PEP

Example: Analysis of Newton method

$$\begin{aligned}
 & \max_{x_k \in \mathbb{R}^d, g_k \in \mathbb{R}^d, h_k \in \mathbb{R}^{d \times d}, p_k \in \mathbb{R}^d} ||x_N - x^*||^2 \\
 & \text{s.t. } \exists f \in \mathcal{F} \text{ s.t. } f(x_k) = f_k, \nabla f(x_k) = g_k, \nabla^2 f(x_k) = h_k, \\
 & \text{(Newton step) } x_{k+1} = x_k - p_k, \\
 & h_k p_k = g_k, \\
 & ||x_0 - x^*||^2 \leq R^2, \\
 & ||g^*||^2 = 0,
 \end{aligned}$$

Or any other second order scheme:

- Cubic Newton method : $T_M(x) \in \text{Arg min}_y \left[\langle f'(x), y-x \rangle + \frac{1}{2} \langle f''(x)(y-x), y-x \rangle + \frac{M}{6} \|y-x\|^3 \right], \quad (2.4)$
[Nesterov, Polyak 2008]
- Damped Newton method: $x_{k+1} = x_k - \frac{1}{1+M_f \lambda_f(x_k)} [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$
- Gradient Regularized Newton method: $\lambda_k = \sqrt{H \|\nabla f(x^k)\|}$
[Mishchenko 2022] $x^{k+1} = x^k - (\nabla^2 f(x^k) + \lambda_k \mathbf{I})^{-1} \nabla f(x^k)$

Interpolation conditions for univariate Hessian Lipschitz functions

We focus on univariate functions for simplicity:

$\mathcal{D}_M$: univariate functions with Lipschitz continuous Hessian.

Step 1: Inequalities
from definition of $\mathcal{F}$

Definition. $f \in \mathcal{D}_M$ if, and only if

$$|f''(x) - f''(y)| \leq M|x - y| \quad \forall x, y. \quad (\text{S})$$

Theorem. If $f \in \mathcal{D}_M$ then,

Not interpolation condition

$$|f(y) - f(x) - f'(x)(y - x) - \frac{1}{2}f''(x)(y - x)^2| \leq \frac{M}{6}|y - x|^3 \quad \forall x, y. \quad (\text{S2})$$

Theorem. $f \in \mathcal{D}_M$ if, and only if

Interpolation condition

$$\begin{aligned} f(y) - f(x) - f'(x)(y - x) - \frac{1}{2}f''(x)(y - x)^2 &\leq \frac{M}{6}|y - x|^3 \\ &- \frac{(f'(y) - f'(x) - f''(x)(y - x) - \frac{M}{2}(y - x)|y - x|)^2}{2(M|y - x| - (f''(y) - f''(x)))} \\ &- \frac{(M|y - x| - (f''(y) - f''(x)))^3}{96M^2} \quad \forall x, y. \end{aligned} \quad (\text{S3})$$

Curiosity: « Does (S2) $\Rightarrow$ (S) ? » is an open question as far as we know

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1. Performance Estimation Problem (PEP) Framework
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Global convergence rate of Cubic Newton Method (1)

$$x_{i+1} = \arg \min_x f(x) + f'(x_i)(x - x_i) + \frac{1}{2}f''(x_i)(x - x_i)^2 + \frac{M}{6}|x - x_i|^3. \quad (\text{CNM})$$

Theorem. (CNM) on Hessian M -Lipschitz univariate functions satisfy

$$f(x_i) - f(x_{i+1}) \geq \frac{M}{12} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2}.$$

Proof.

$$\begin{aligned} f(x_{i+1}) &\leq f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{1}{2}f''(x_i)(x_{i+1} - x_i)^2 + \frac{M}{6}|x_{i+1} - x_i|^3 \\ &\leq \dots \\ &\leq f(x_i) - \frac{M}{12} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2} \end{aligned}$$

[Nesterov, Polyak 2008] (in multivariate)

□

Global convergence rate of Cubic Newton Method (1)

$$x_{i+1} = \arg \min_x f(x) + f'(x_i)(x - x_i) + \frac{1}{2}f''(x_i)(x - x_i)^2 + \frac{M}{6}|x - x_i|^3. \quad (\text{CNM})$$

Theorem. (CNM) on Hessian M -Lipschitz univariate functions satisfy

$$f(x_i) - f(x_{i+1}) \geq \frac{5M}{12} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2}.$$

Proof.

$$\begin{aligned} f(x_{i+1}) &\leq f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{1}{2}f''(x_i)(x_{i+1} - x_i)^2 + \frac{M}{6}|x_{i+1} - x_i|^3 - P_1(x_i, x_{i+1}) - P_2(x_i, x_{i+1}) \\ &\leq \dots \\ &\leq f(x_i) - \frac{M}{12} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2} - \frac{M}{3} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2} \\ &\leq f(x_i) - \frac{5M}{12} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2} \end{aligned}$$

Step 1: Inequalities
from definition of $\mathcal{F}$

[Nesterov, Polyak 2008] (in multivariate)

[Rubbens, B, Hendrickx, Glineur 2024] $\square$

Global convergence rate of Cubic Newton Method (2)

$$x_{i+1} = \arg \min_x f(x) + f'(x_i)(x - x_i) + \frac{1}{2}f''(x_i)(x - x_i)^2 + \frac{M}{6}|x - x_i|^3. \quad (\text{CNM})$$

Theorem. (CNM) on Hessian M -Lipschitz univariate functions satisfy

$$f(x_i) - f(x_{i+1}) \geq \frac{M}{12} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2}.$$

Moreover, if the function is bounded below by f^* , then

[Nesterov, Polyak 2008] (in multivariate)
$$\min_{i=1,\dots,N} |f'(x_i)| \leq 4M \left(\frac{3(f(x_0) - f^*)}{2MN} \right)^{2/3}.$$

Step 1: Inequalities
from definition of $\mathcal{F}$

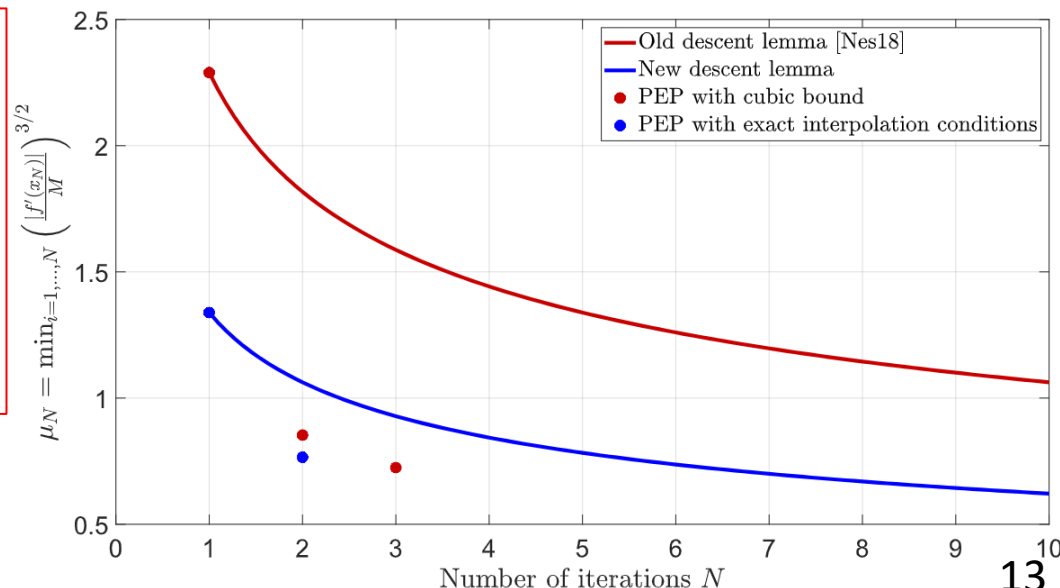
Step 2: combination
of inequalities and
iteration of $\mathcal{M}$

Theorem. (CNM) on Hessian M -Lipschitz univariate functions satisfy

$$f(x_i) - f(x_{i+1}) \geq \frac{5M}{12} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2}.$$

Moreover, if the function is bounded below by f^* , then

[Rubbens, B, Hendrickx, Glineur 2024]
$$\min_{i=1,\dots,N} |f'(x_i)| \leq \frac{4M}{5^{2/3}} \left(\frac{3(f(x_0) - f^*)}{2MN} \right)^{2/3}.$$



Local quadratic convergence rate of Newton Method

Theorem. *If*

- *f has a M -Lipschitz continuous Hessian,*
- *$\exists x^*$ such that $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) = \mu I \succ 0$,*
- *$\frac{M}{\mu} \|x_0 - x^*\| \leq \frac{2}{3}$,*

then all Newton iterations $x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k)$ satisfy

$$\|x_{k+1} - x^*\| \leq \frac{\frac{M}{\mu} \|x_k - x^*\|^2}{2 \left(1 - \frac{M}{\mu} \|x_k - x^*\|\right)}$$

[Nesterov 2018]

Observation: PEP numerical results exactly match the bound

Theorem. *Theorem above is tight and attained by the following univariate cubic by parts function.*

$$f_1(x) = \begin{cases} \frac{Mx^3}{6} + \mu \frac{x^2}{2} & \text{if } x \leq 0, \\ -\frac{Mx^3}{6} + \mu \frac{x^2}{2} & \text{if } x > 0. \end{cases}$$

[Rubbens, B, Hendrickx, Glineur 2024]

Univariate case is « sufficiently rich » to attain the worst-case performance

Optimal step size of fixed damped Newton method

Fixed Damped Newton method : $x_{k+1} = x_k - \alpha \frac{f'(x_k)}{f''(x_k)}$

α that optimize the worst-case performance?

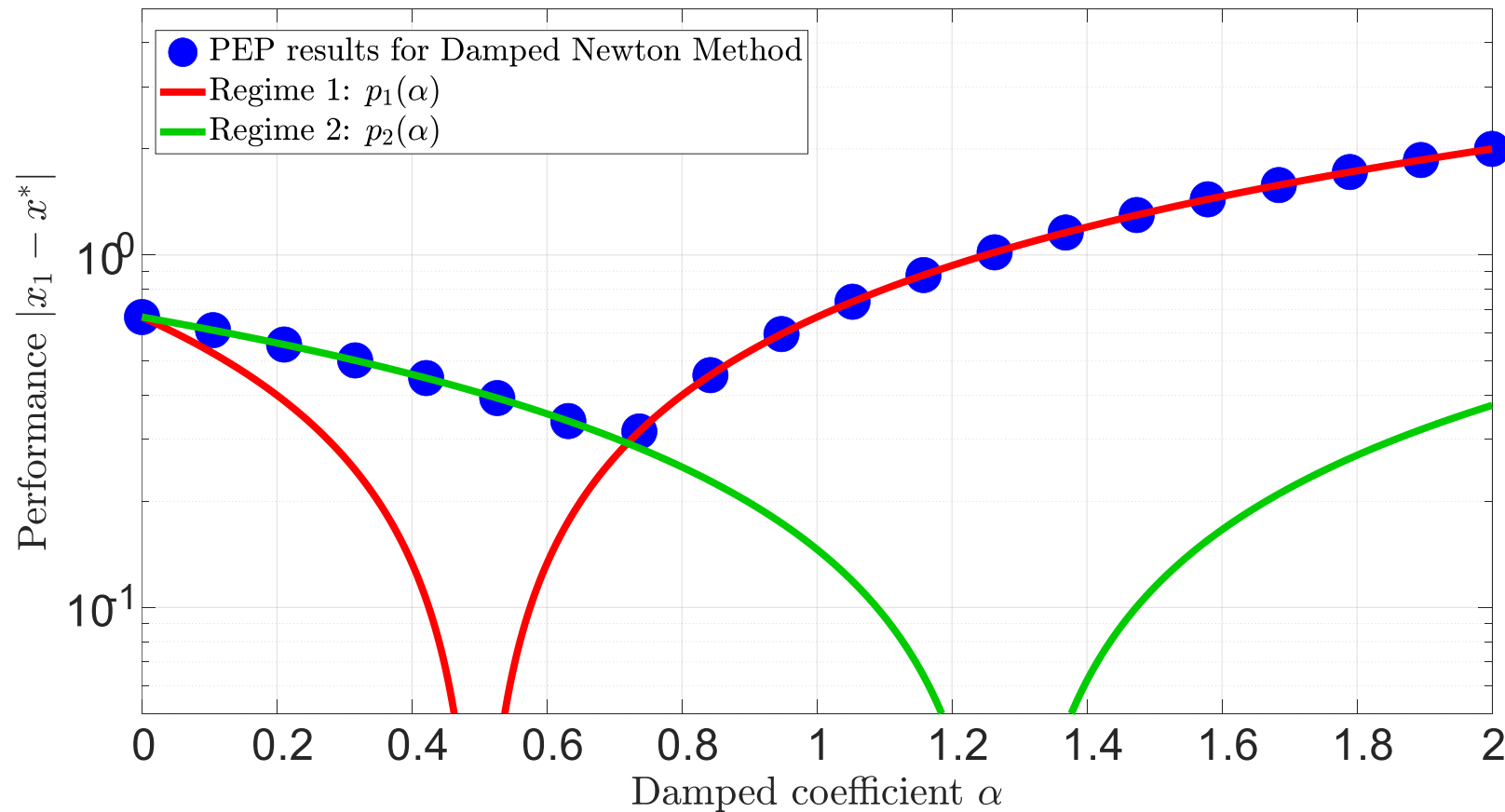


Fig. 1: $M = \mu = 1$ and $|x_0 - x^*| = \frac{2}{3}$

Summary (1/2)

State of the art

1. Tight worst-case performance requires **Step 1: Inequalities** and **Step 2: Combination of them**
2. PEP combines them **automatically and optimally**
3. Convex PEP is very efficient and useful to analyze **fixed first-order methods** (see **PEPit's documentation**)
4. Non-convex PEP allows to **analyze any method but is very costly**

Contributions

1. Interpolation conditions for **univariate Hessian Lipschitz functions**

Theorem. $f \in \mathcal{D}_M$ if, and only if

$$\begin{aligned} f(y) - f(x) - f'(x)(y-x) - \frac{1}{2}f''(x)(y-x)^2 &\leq \frac{M}{6}|y-x|^3 \\ &\quad - \frac{(f'(y) - f'(x) - f''(x)(y-x) - \frac{M}{2}(y-x)|y-x|)^2}{2(M|y-x| - (f''(y) - f''(x)))} \\ &\quad - \frac{(M|y-x| - (f''(y) - f''(x)))^3}{96M^2} \quad \forall x, y. \end{aligned}$$

2. Applying non-convex PEP to **second-order methods**

Summary (2/2)

Contributions

3. Improved Descent lemma of CNM by a factor 5 (for univariate functions)

Theorem. (CNM) on Hessian M -Lipschitz univariate functions satisfy

$$f(x_i) - f(x_{i+1}) \geq \frac{5M}{12} \left(\frac{|f'(x_{i+1})|}{M} \right)^{3/2}.$$

4. Exhibit a function attaining the worst local quadratic convergence of Newton method

5. Step size selection of damped Newton method (for univariate functions)

$$f_1(x) = \begin{cases} \frac{Mx^3}{6} + \mu \frac{x^2}{2} & \text{if } x \leq 0, \\ -\frac{Mx^3}{6} + \mu \frac{x^2}{2} & \text{if } x > 0. \end{cases}$$

Future perspectives

Exploiting **non-convex PEP** to analyze new:

1. **Second-order schemes:** Gradient regularized Newton method, adaptive damped Newton method, etc
2. **Classes of functions:** self-concordant, etc
3. **Optimization methods:** zeroth order, adaptive, quasi-Newton methods, etc